Local Normal Forms for Complex Base Expansions: A General (s, b, l) Framework and the φ Specialization

VR & AI

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Abstract

We present a local, phase-aware normalization theory for complex base expansions built from three parameters: sites per turn $l \in \mathbb{N}_{\geq 2}$, per-step radial growth $b > 1$, and a per-level shrinkage factor $s \in (0, 1]$. For canonical integer-digit normalization we assume b is a quadratic algebraic integer. We prove a *tri-site, same-phase* carry identity that yields termination and confluence on each residue class ("phase track"), provide digit-set choices and carry-count bounds, and show how to obtain global self-similarity with track-dependent angular lattices. We then specialize to φ, recovering Lucas-number coefficients and the familiar ± 1 neighbor effects.

1 Setup: the general (s, b, l) model

Fix integers $l \geq 2$ and parameters $b > 1$, $s \in (0, 1]$. Let

$$U := e^{i\,2\pi/l} \qquad \text{and} \qquad B := b\,U.$$

Definition 1.1 (Complex (b, l)-string; phase tracks)**.** A complex (b, l)-string is an integer sequence $(d_k)_{k\in\mathbb{Z}}$ satisfying:

1. *finite right tail:* $\exists K \in \mathbb{Z}$ such that $d_k = 0$ for all $k > K$;
2. *absolutely summable left tail:* $\sum_{t\geq 0} |d_{-t}|\, b^{-t} < \infty$.

Its value is the absolutely convergent series

$$\mathrm{Val}(d) := \sum_{k\in\mathbb{Z}} d_k\, B^k, \qquad \text{with } |B| = b.$$

Indices decompose into l *phase tracks* by residue $r \equiv k \pmod{l}$, where $k = r + lt$ and $t \in \mathbb{Z}$. On a fixed track r we write $d_t := d_{r+lt}$.

Definition 1.2 (Hierarchical sum with shrink s)**.** For levels $\ell = 0, 1, \ldots,$ a finite-depth address chooses integers $d_{\ell,k}$ and positions k_ℓ; its value is

$$z \;=\; \sum_{\ell\geq 0} s^\ell \sum_{k\in\mathbb{Z}} d_{\ell,k}\, B^k.$$

For infinite depth, $s^\ell \to 0$ and absolute convergence of each inner sum ensure z is well-defined.

Remark 1.3 (Quadratic b for integer carries). To obtain *integer* carry coefficients (and thus integer digits under normalization), we assume b is a root of

$$b^2 - P\,b + Q \;=\; 0, \qquad P, Q \in \mathbb{Z}, \quad b > 1. \tag{1}$$

This includes all metallic means ($Q = -1$, $P \geq 1$) and their integer powers b^h.

Convergence of hierarchical sums (sufficient bound)

Let $\beta_l := b^l > 1$. If, for each level ℓ, the per-track reindexed series obeys

$$\sum_{t\in\mathbb{Z}} |d_{\ell,r+lt}|\,\beta_l^t \;\le\; C_\ell \quad \text{for all residues } r,$$

then

$$|z| \;\le\; \sum_{\ell\ge 0} s^\ell \sum_{r=0}^{l-1} C_\ell.$$

Thus $s<1$ and $\sup_\ell C_\ell < \infty$ imply absolute convergence.

Post-normalization quantification. After per-track normalization into a digit set of size $V := V_l(P,Q)$, each digit satisfies $|d_{\ell,r+lt}| \le V-1$ (or $\le a = \lfloor (V-1)/2 \rfloor$ in the balanced case). If the normalized support on level ℓ has ν_ℓ nonzero sites in total and the maximal occupied site index is $t_{\max,\ell}$ (measured on its track), then the crude but effective bound

$$\boxed{C_\ell \;\le\; (V-1)\,\beta_l^{t_{\max,\ell}}\,\nu_\ell}$$

holds, since $\beta_l^t \le \beta_l^{t_{\max,\ell}}$ for all occupied t. When occupied sites form blocks, the sharper geometric sum $\sum_{t=a}^{b} \beta_l^t = \beta_l^a \frac{\beta_l^{b-a+1}-1}{\beta_l-1}$ may be used trackwise; either way C_ℓ scales linearly in the (finite) active support at level ℓ.

Numerical illustration (φ, $l=4$). Here $\beta_l = \varphi^4 \approx 6.854$. Suppose a level ℓ has $\nu_\ell = 3$ occupied sites on a track, at $t \in \{-1,0,1\}$, with digits bounded by $V-1 = L_4 - 1 = 6$. The crude bound yields

$$C_\ell \;\le\; (V-1)\,\beta_l^{t_{\max,\ell}}\,\nu_\ell \;=\; 6\cdot 6.854^{\,1}\cdot 3 \;\approx\; 123.4.$$

Using the geometric sum per block (here contiguous from -1 to 1):

$$\sum_{t=-1}^{1} \beta_l^t \;=\; \beta_l^{-1}\,\frac{\beta_l^3-1}{\beta_l-1} \;\approx\; 0.146\cdot\frac{321.996-1}{5.854} \;\approx\; 8.0,$$

so $C_\ell \le 6\cdot 8.0 \approx 48$, a tighter bound.

2 Local identity and normalization on one lattice (b,l)

Let $U_n(P,Q), V_n(P,Q)$ be the Lucas sequences:

$$U_0 = 0,\, U_1 = 1, \quad V_0 = 2,\, V_1 = P, \quad X_{n+1} = PX_n - QX_{n-1}.$$

From (1) one has for $n \ge 1$:

$$b^{2n} = V_n(P,Q)\,b^n - Q^n. \tag{2}$$

Proposition 2.1 (Phase-aligned (tri-site) carry identity)**.** *For all $k \equiv r \pmod{l}$,*

$$B^{k+2l} - V_l(P,Q)\,B^{k+l} + Q^l\,B^k \;=\; 0. \tag{3}$$

Equivalently on a track (with $\beta_l = b^l > 1$):

$$\beta_l^{t+1} \;=\; V_l(P,Q)\,\beta_l^t \;-\; Q^l\,\beta_l^{t-1}.$$

Proof of Proposition 2.1. Let b be quadratic with $b^2 - Pb + Q = 0$, and let $U = e^{i2\pi/l}$. The Lucas identity $b^{2n} = V_n(P,Q)\, b^n - Q^n$ holds for all $n \geq 1$. Taking $n = l$ gives $b^{2l} = V_l(P,Q)\, b^l - Q^l$. Multiply by U^k and note $U^l = 1$:

$$(bU)^{k+2l} - V_l(P,Q)\,(bU)^{k+l} + Q^l\,(bU)^k \;=\; 0.$$

Since $B = bU$, this is exactly $B^{k+2l} - V_l(P,Q)\,B^{k+l} + Q^l B^k = 0$, and it holds for every $k \equiv r \pmod{l}$, i.e. along each residue track. □

Local moves (per track). For $t^\star \in \mathbb{N}$, define the value-preserving carry/borrow at center t by

$$\begin{aligned} \textbf{carry:}\ & (d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} + Q^l t^\star,\ d_t - V_l(P,Q)t^\star,\ d_{t+1} + t^\star), \\ \textbf{borrow:}\ & (d_{t-1}, d_t, d_{t+1}) \mapsto (d_{t-1} - Q^l t^\star,\ d_t + V_l(P,Q)t^\star,\ d_{t+1} - t^\star). \end{aligned} \tag{4}$$

Definition 2.2 (Digit sets)**.** Normalize per track into a contiguous set $\mathcal{D}$ of size $V_l(P,Q)$:

$$\mathcal{D} \;=\; \begin{cases} \{0, 1, \ldots, V_l(P,Q) - 1\}, & \text{nonnegative}, \\ \{-a, \ldots, a\},\ a = \lfloor (V_l(P,Q) - 1)/2 \rfloor, & \text{balanced}. \end{cases}$$

Algorithm 1: Deterministic per-track normalization

Let $V := V_l(P,Q)$ and let the target digit set be either $\{0, \ldots, V-1\}$ (nonnegative) or the balanced set $\{-a, \ldots, a\}$ with $a = \lfloor (V-1)/2 \rfloor$. The following rightmost-first procedure yields the canonical normal form guaranteed by Theorem 2.3.

Input: integer array $(d_t)_{t \in \mathbb{Z}}$ with finite right tail. **Output:** normalized digits $(\hat{d}_t)$ in $\mathcal{D}$.

1. **Rightmost selection.** While some t has $d_t \notin \mathcal{D}$, choose the *maximum* such index $t^\star$.
2. **One-shot fix at $t^\star$.**
   - *Nonnegative digits:* If $d_{t^\star} \geq V$, set $t = \lfloor d_{t^\star}/V \rfloor$ and apply a carry with multiplicity t:

     $$(d_{t^\star-1}, d_{t^\star}, d_{t^\star+1}) \leftarrow (d_{t^\star-1} + Q^l t,\ d_{t^\star} - Vt,\ d_{t^\star+1} + t).$$

     If $d_{t^\star} < 0$, set $t = \lceil (-d_{t^\star})/V \rceil$ and apply a borrow of multiplicity t:

     $$(d_{t^\star-1}, d_{t^\star}, d_{t^\star+1}) \leftarrow (d_{t^\star-1} - Q^l t,\ d_{t^\star} + Vt,\ d_{t^\star+1} - t).$$

   - *Balanced digits:* Replace the division by V with centered rounding $t = \lfloor (d_{t^\star} + a)/V \rfloor$ so that the new center lands in $\{-a, \ldots, a\}$ in one step.
3. **Repeat** Step 1.

Theorem 2.3 (Termination and confluence (single lattice))**.** *Fix a track and $q \in (1, \beta_l)$ with $\beta_l = b^l$. The potential $E_q = \sum_{t \in \mathbb{Z}} |d_t| q^t$ strictly decreases under one elementary carry (4). Hence normalization terminates on the track. Overlaps of carries occur only at adjacent centers; critical pairs are joinable. Summing the l tracks yields a unique normal form for the string.*

Termination. A carry at t changes

$$\Delta E_q \;\leq\; q^{t-1}\big(|Q|^l - V_l(P,Q)\,q + q^2\big).$$

The quadratic has roots $q_\pm = b^l$ and $|Q|^l/b^l$. For any $q \in (1, b^l)$ we have $\Delta E_q < 0$. □

Lemma 2.4 (Critical-pair check (adjacent carries commute))**.** *Let C_t denote a unit carry at center t (i.e. $t^\star = 1$), and write $V := V_l(P, Q)$, $Q_0 := Q^l$. Restrict to $\{t-1, t, t+1, t+2\}$ with digits (a, b, c, d). Then*

$$C_{t+1}\, C_t(a, b, c, d) \;=\; C_t\, C_{t+1}(a, b, c, d) \;=\; (a + Q_0,\; b - V + Q_0,\; c + 1 - V,\; d + 1).$$

By linearity, carries with $t^\star > 1$ decompose into sums of unit carries. Hence overlapping carries at t and $t{+}1$ commute.

Proof. Apply the updates in both orders and compare; the two linear maps coincide on (a, b, c, d). □

Confluence. Local confluence follows from commutation of adjacent carries and disjointness of nonadjacent ones. With termination, Newman's lemma yields global confluence and a unique per-track normal form. □

Tick and cost. A *tick* adds $+B^0$ (track $r = 0$, site $t = 0$) and normalizes along that track. If the saturated block length is n, the tick uses $O(n)$ carries.

Numerical example (φ, $l = 4$)

Here $P = 1$, $Q = -1$, $V_l = V_4 = 7$, $Q^l = +1$, digit set $\mathcal{D} = \{0, 1, \ldots, 6\}$. On a single track (say $r = 0$), start with $(d_{-1}, d_0, d_1) = (0, 13, 0)$. One carry at $t = 0$ gives $(1, 6, 1)$, already normalized; value is preserved because $B^{k+8} - 7B^{k+4} + B^k = 0$.

Adjacent multi-carry example ($\varphi, l = 4$)

Here $P{=}1$, $Q{=}-1$, hence $V = 7$ and $Q^l = +1$. Consider a window $(d_{t-1}, d_t, d_{t+1}, d_{t+2}) = (0, 13, 12, 0)$. Both centers t and $t{+}1$ require carries.
Order A: C_t then C_{t+1}.

$$(0, 13, 12, 0) \xrightarrow{C_t} (1, 6, 13, 0) \xrightarrow{C_{t+1}} (1, 7, 6, 1) \xrightarrow{C_t} (2, 0, 7, 1) \xrightarrow{C_{t+1}} (2, 1, 0, 2).$$

Order B: C_{t+1} then C_t.

$$(0, 13, 12, 0) \xrightarrow{C_{t+1}} (0, 14, 5, 1) \xrightarrow{C_t} (1, 7, 6, 1) \xrightarrow{C_t} (2, 0, 7, 1) \xrightarrow{C_{t+1}} (2, 1, 0, 2).$$

Both orders terminate at the same normalized state, illustrating Lemma 2.4.

3 Digit-set variants and fast addition

- **Balanced signed digits.** With $\mathcal{D} = \{-a, \ldots, a\}$, a provisional sum s_t satisfies $|s_t| \le 2a \le V_l - 1$. A *single* carry at t reduces $|s_t|$ to $\le a$; this yields one-pass, bounded-carry addition per track (plus bounded neighbor fixes).
- **Sparse (NAF-like).** Use a small signed set (e.g. $\{-1, 0, +1\}$) and forbid adjacent nonzeros on a track. Add a local penalty to the potential to keep termination; sparsity reduces Hamming weight.

4 Carry-count bounds (per track)

Let $S = \sum_t |d_t|$.

Proposition 4.1 (Uniform ℓ^1 bound)**.** *Assume $|Q| \le 1$ (metallic and similar). Each carry reduces the center by $V_l(P,Q)$ and increases the two neighbors' absolute sum by at most $|Q|^l+1 \le 2$. Hence*

$$\#\text{carries} \;\le\; \frac{S}{V_l(P,Q) - (|Q|^l + 1)}.$$

Metallic case: *when $|Q| = 1$ this simplifies to $S/(V_l(P,Q) - 2)$.*

Proposition 4.2 (Weighted bound)**.** *For $q \in (1, b^l)$ and $E_q = \sum_t |d_t| q^t$, the choice $q = V_l(P,Q)/2 < b^l$ gives*

$$\Delta E_q \;\le\; -\Big(\tfrac{V_l(P,Q)^2}{4} - |Q|^l\Big)\, q^{t-1},$$

hence

$$\#\text{carries} \;\le\; \frac{E_q^{\text{init}}}{\frac{V_l(P,Q)^2}{4} - |Q|^l}.$$

Edge case check: *for φ with $l = 2$, $V_l = 3$ and $b^l = \varphi^2 \approx 2.618 > 3/2$, so $q = V_l/2$ is admissible.*

Carry-count instantiation (φ, $l = 4$). On one track with initial digits $(0, 13, 12, 0)$ we have $S = 25$, $V = 7$, $|Q|^l = 1$. The uniform ℓ^1 bound gives $\# \le 25/(7-2) = 5$. With $q = 3.5$, $E_q = 13 + 12q = 55$, so the weighted bound gives $\# \le 55/(49/4 - 1) \approx 4.89$. A concrete normalization needs exactly 4 carries (cf. the example above), showing both bounds are nontrivial and reasonably tight.

5 Track-dependent l_r with global self-similarity

Fix a *global similarity exponent* $M \in \mathbb{N}$ and set $\beta := b^M > 1$. Choose tracks r; for each track pick $l_r \mid M$ and define

$$h_r := \frac{M}{l_r} \in \mathbb{N}, \qquad B_r := b^{h_r}\, e^{i2\pi/l_r}.$$

Then $(b^{h_r})^{l_r} = \beta$ (same full-turn radial factor on all tracks). With level shrink s^ℓ, the value is

$$z = \sum_{\ell \ge 0} s^\ell \sum_r \sum_{k \in \mathbb{Z}} d_{r,\ell,k}\, B_r^k.$$

Proposition 5.1 (Uniform carry across tracks)**.** *If b satisfies $x^2 - Px + Q = 0$, then for each track r and $k \equiv r \pmod{l_r}$,*

$$B_r^{k+2l_r} - V_M(P,Q)\, B_r^{k+l_r} + Q^M\, B_r^k = 0. \tag{5}$$

Thus every track uses the same *coefficients $V_M(P,Q)$ and Q^M; the digit-set size is $V_M(P,Q)$. Termination/confluence and bounds hold uniformly with $\beta = b^M$.*

Proof of Proposition 5.1. Fix the global $M \in \mathbb{N}$ and a track with $l_r \mid M$, $h_r = M/l_r$. Then b^{h_r} is quadratic over $\mathbb{Z}$ with the same (P,Q), hence

$$(b^{h_r})^{2l_r} = V_{l_r}(P,Q)\,(b^{h_r})^{l_r} - Q^{l_r}.$$

Because $h_r l_r = M$, the composed Lucas coefficient is $V_{h_r l_r}(P,Q) = V_M(P,Q)$ and $Q^{h_r l_r} = Q^M$. Thus $b^{2M} = V_M(P,Q)\, b^M - Q^M$. Multiplying by U_r^k with $U_r^{l_r} = 1$ yields

$$B_r^{k+2l_r} - V_M(P,Q)\, B_r^{k+l_r} + Q^M B_r^k = 0,$$

which is track-independent. □

Notation. We write $\beta_l := b^l > 1$ for a single lattice's full-turn factor and $\beta := b^M > 1$ for the common factor in the multi-track construction $(l_r \mid M)$. Choosing $s = \beta^{-\nu}$ ($\nu \in \mathbb{N}$) makes the level-ℓ pattern a scaled copy of level $\ell - \nu$. One full turn on any track scales by β.

6 Specialization to φ

Set $b = \varphi = \frac{1+\sqrt{5}}{2}$, which satisfies $x^2 - x - 1 = 0$ ($P = 1$, $Q = -1$). Then

$$V_n(P, Q) = L_n \quad \text{(Lucas numbers)}, \qquad Q^n = (-1)^n.$$

Single lattice (φ, l). The carry identity becomes

$$B^{k+2l} - L_l\, B^{k+l} + (-1)^l B^k = 0, \qquad B = \varphi\, e^{i2\pi/l},$$

and per-track digits normalize into a set of size L_l; with l even, $\{0, \ldots, L_l - 1\}$ is natural. The potential window is $(1, \varphi^l)$.

Example 6.1 (φ and quarter-turns). Take $l = 4$. Then $L_4 = 7$ and $B^{k+8} - 7B^{k+4} + B^k = 0$ ($(-1)^4 = +1$). Nonzeros, e.g. at $k \in \{-2, 2, 5\}$, decompose by residues: $r = 2$ has $t = -1$ and $t = 0$; $r = 1$ has $t = 1$; tracks $r = 0, 3$ empty. Since digits lie in $\{0, \ldots, 6\}$, no carries are needed here.

7 Representation properties (scope and limits)

Our canonical normalization provides a unique normal form *within the language* of (b, l)-strings with finite right tail and absolutely summable left tail. Two classical questions are (i) *existence/completeness* (does every $z \in \mathbb{C}$ admit such an expansion?) and (ii) *uniqueness before normalization.*

Existence (scope). With the finite-right constraint, the value set is

$$\mathcal{V} = \left\{ \sum_{k \in \mathbb{Z}} d_k B^k : \text{ finite right tail, absolutely summable left tail, } d_k \in \mathbb{Z} \right\}.$$

It is closed under addition and under multiplication by $B^{\pm 1}$, but it is not all of $\mathbb{C}$. For quadratic algebraic b and sufficiently rich digit sets, existence results for *canonical number systems* are known in related settings (e.g. Gaussian and imaginary quadratic bases [7, 3, 5, 1]). Our framework is compatible with those existence theories but we do not claim completeness for $\mathbb{C}$ under the present (finite-right) constraint.

Uniqueness. Pre-normalization representations can be nonunique in complex bases (see [4]); our tri-site calculus yields a *canonical* representative per residue track. Thus normalization provides uniqueness *modulo* the local identities we allow, i.e. within the congruence generated by (3) (and (5) in the multi-track case).

Density. Allowing infinite left tails makes $\mathcal{V}$ infinite and "one-sided dense" along rays, but not dense in $\mathbb{C}$ under the finite-right restriction. Extending the framework to two-sided infinite strings (outside the present scope) would address density/completeness questions but requires additional regularity to retain uniqueness.

8 Figures

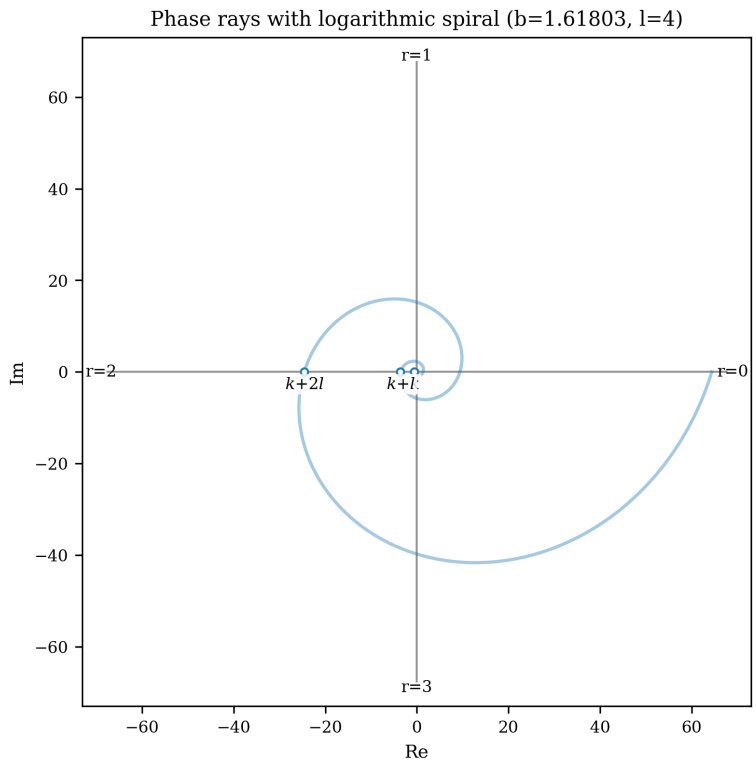

Figure 1: **Tri-site normalization is ray-local.** Example with $b = \varphi$, $l = 4$. The discrete sites lie on a logarithmic spiral (faint curve) with per-step growth b and rotation $2\pi/l$. Residue classes $r = 0, 1, 2, 3$ appear as fixed rays (same angle, every full turn). A tri-site triple $k, k{+}l, k{+}2l$ (all on the same ray) is highlighted; the carry identity $B^{k+8} - 7B^{k+4} + B^k = 0$ acts only on these three collinear sites, preserving value.

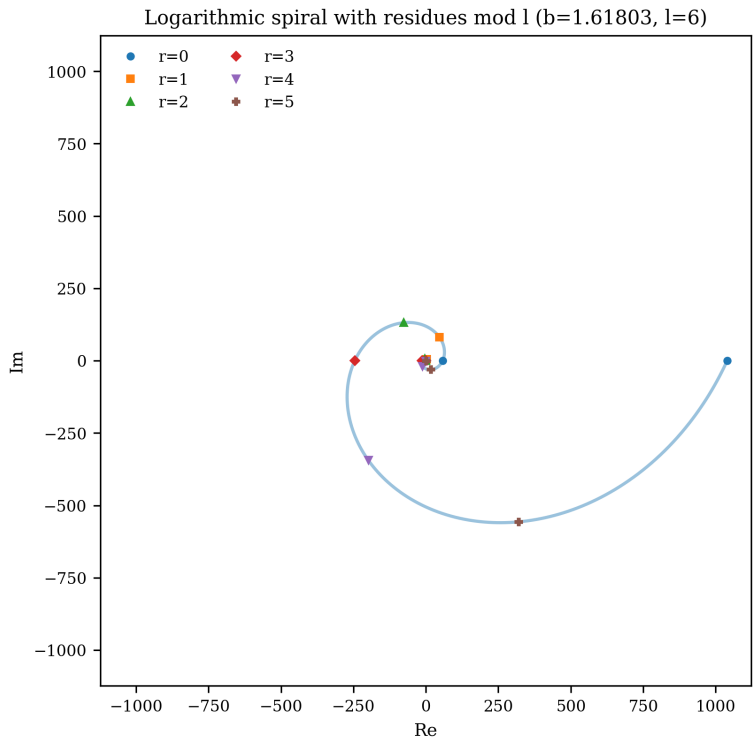

Figure 2: **Residues indicated on a logarithmic spiral.** Example with $b = \varphi$, $l = 6$. Consecutive sites step along the spiral; markers distinguish residues $k \bmod l$. Within a fixed residue r, indices differ by multiples of l and align radially, forming a phase track.

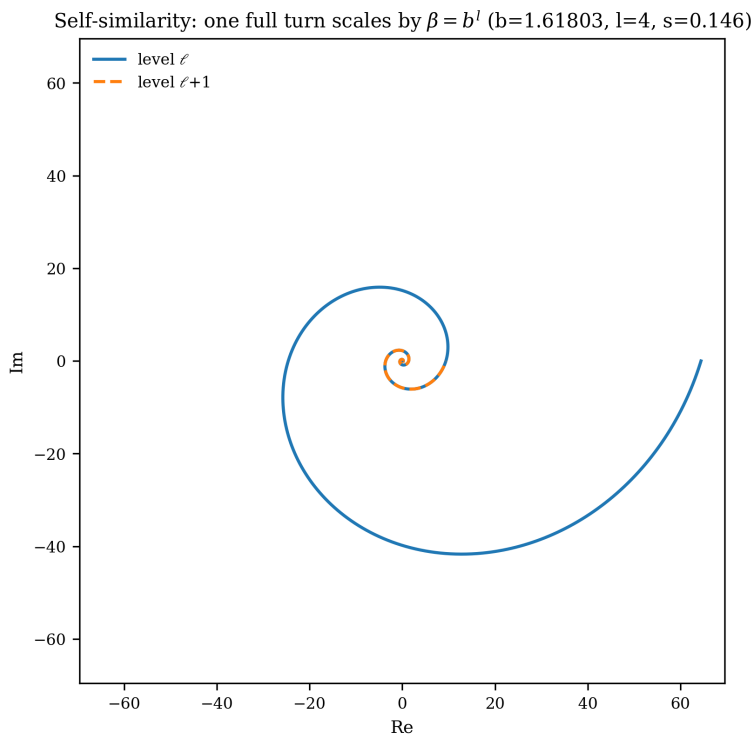

Figure 3: **Level self-similarity.** Two spirals overlaid for levels ℓ (solid) and ℓ+1 (dashed) with $b = \varphi$, $l = 4$, and $s = \beta_l^{-\nu}$ for an integer $\nu \geq 1$. One full turn on any phase track multiplies radius by $\beta_l = b^l$; choosing $s = \beta_l^{-\nu}$ yields exact self-similarity after ν turns.

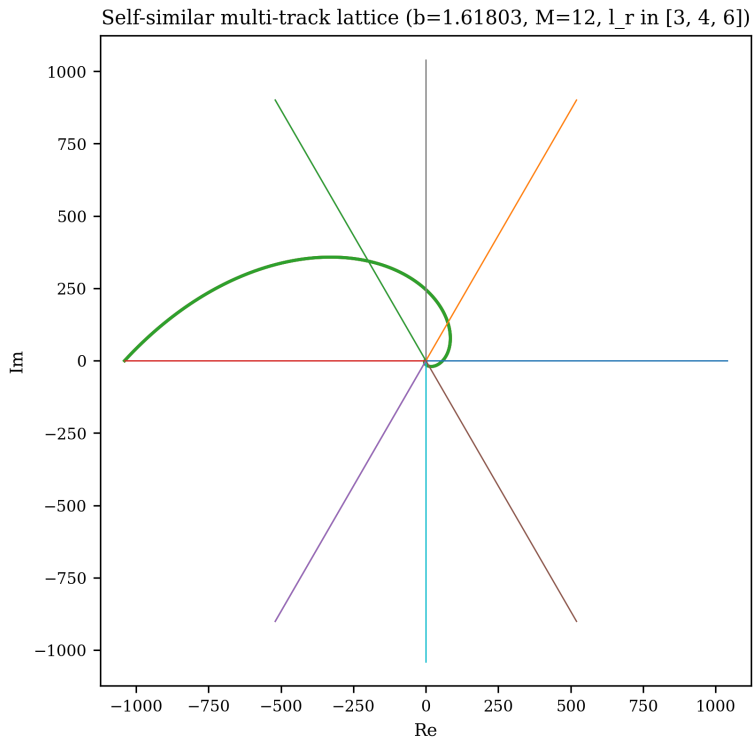

Figure 4: **Track-dependent angular lattices with common full-turn factor.** Example with global similarity exponent $M = 12$, tracks $l_r \in \{3, 4, 6\}$, and $b = \varphi$. Each track uses base $B_r = \varphi^{M/l_r} e^{i2\pi/l_r}$, so one full turn scales radius by the common factor $\beta = b^M = \varphi^{12}$. The per-level shrink s (not shown) can be chosen as $s = \beta^{-1}$ to obtain exact self-similarity level-to-level.

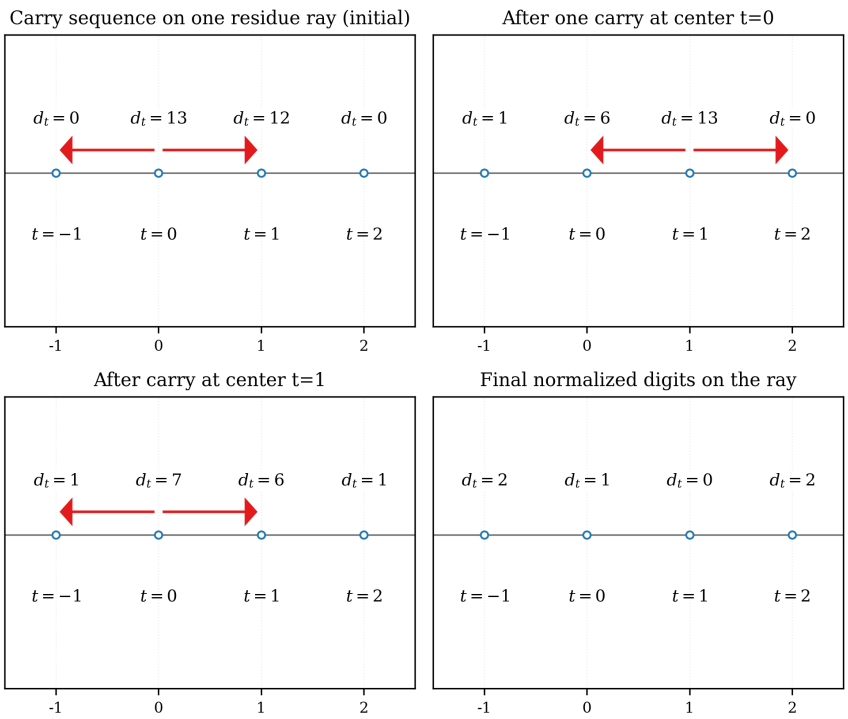

Figure 5: **Carry sequence (strip view) on one residue ray** for $(\varphi, l{=}4)$, residue $r{=}0$. Sites $t \in \{-1, 0, 1, 2\}$ are arranged along a 1D strip; digits d_t are annotated above each site and arrows indicate carries. This overlap-free layout shows the sequence at a glance without geometric clutter.

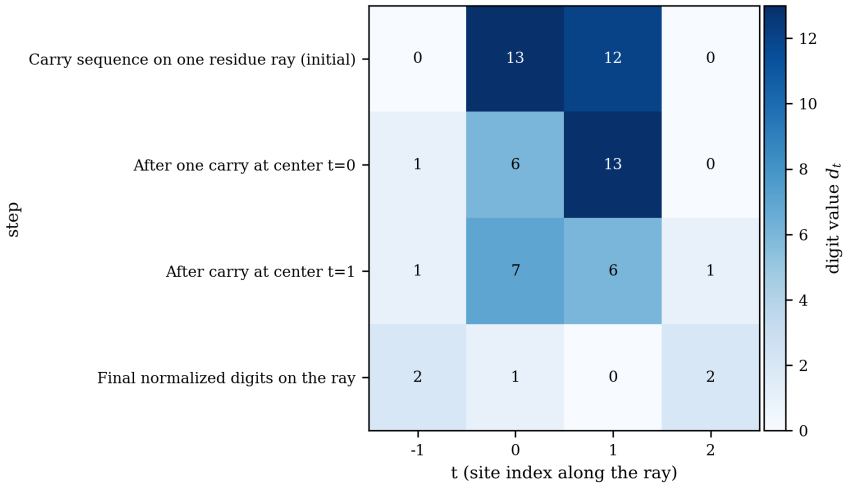

Figure 6: **Carry sequence (matrix view).** Rows correspond to steps in the sequence; columns correspond to sites t. Each cell shows the digit value d_t (color and numeric annotation). This compact panel highlights how adjacent carries commute to yield the same final normal form $(2, 1, 0, 2)$, illustrating Lemma 2.4.

9 Related work (brief pointers)

Complex bases and non-standard radices have a rich history. Knuth [6] discusses negative and unusual bases in a foundational algorithmic context. Penney [7] introduced base $-1+i$ arithmetic. Gilbert [3, 4] studied arithmetic and representations in complex bases. Frougny and Sakarovitch [2] surveyed automatic/normalized representations in algebraic bases. Our contribution is a *phase-aware tri-site* normalization that (i) works uniformly on residue tracks, (ii) gives simple termination/confluence proofs via a potential, and (iii) extends cleanly to track-dependent lattices with *uniform* carry coefficients under a global self-similarity constraint (see also [5, 1] for modern developments on canonical number systems in quadratic/imaginary settings). Unlike [5], which establishes canonical number systems in imaginary quadratic fields (existence/tilings for $\mathbb{Z}[\alpha]$), and [1], which surveys CNS criteria across algebraic settings, our focus is on *phase-aware local normalization* with uniform tri-site coefficients per residue track, a tool that applies uniformly across complex rotations $e^{i2\pi/l}$ and supports self-similar, track-dependent lattices.

10 Beyond quadratic bases (remark)

If b has minimal polynomial of degree $d > 2$, the analogue of (3) is a local identity with at most $(d+1)$ sites on each residue track, obtained by multiplying the minimal polynomial by U^k and grouping indices by steps of l. The stencil length equals the number of nonzero coefficients of the polynomial. *Example (cubic).* If b is the real root of $x^3 - 2x - 1 = 0$, then $b^3 = 2b + 1$, so for any $l \geq 2$,

$$B^{k+3l} - 2\,B^{k+l} - B^k = 0, \qquad B = be^{i2\pi/l}.$$

This is a tri-site rule with larger spacing. The same potential method applies: choose $q \in (1, b^l)$ so that the induced quadratic in q is negative on that interval; termination and confluence then follow as in the quadratic case, with integer coefficients unchanged.

11 Conclusion and outlook

Starting from the minimal (s, b, l) data with b quadratic, we obtain phase-aware, value-preserving local carries that terminate and are confluent, yielding canonical normal forms with explicit complexity bounds. The φ specialization recovers Lucas coefficients and simple signs. A track-dependent choice $l_r \mid M$ produces a self-similar multi-track system with *uniform* normalization across tracks. *Applications:* potential links include efficient on-line arithmetic in nonstandard bases, symbolic dynamics on subshifts with local normalization, and geometric encodings for quasicrystal tilings.

References

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